

# Zhenhao Liu

Undergraduate Researcher in NLP

 [likc1606.github.io](https://github.com/likc1606) •  [github.com/LikC1606](https://github.com/LikC1606)

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Research focus: automated scientific research, scientific agents, and NLP for science

## EDUCATION

Westlake University

Hangzhou, China

Undergraduate, Artificial Intelligence

2024 cohort; GPA 3.753/4.3

**Selected coursework:** Natural Language Processing (A), Machine Learning (A+), Data Structures & Algorithms (A+), Large Language Models (A)

## RESEARCH INTERESTS

**Automated scientific research** scientific agents for literature discovery, hypothesis generation, experiment design, coding, and reproducible evaluation

**NLP for science** data-efficient language modeling, cognitive NLP, and brain-language modeling

## RESEARCH EXPERIENCE

Westlake NLP Lab

Hangzhou, China

[Advisor profile](#)

*Undergraduate Researcher; advised by Prof. Yue Zhang*

- Lead research on data-efficient Chinese language modeling and cognitive NLP for word- and sentence-level fMRI encoding.
- Build reproducible evaluation pipelines with controlled representation comparisons, participant-cluster bootstrap, leave-one-participant-out analyses, and sensitivity checks.

## PUBLICATIONS

RepDuet at NLPCC 2026 Chinese BabyLM: Task-Specific Hybrid Representations for the Cognitive Track

[Model](#)

[Paper PDF](#)

*Sole first author; accepted to NLPCC 2026 Shared Tasks Poster*

- Developed task-specific hybrid representations: intermediate contextual states plus corpus statistics for word-level evaluation, and upper-layer states plus lexical and confidence features for sentence-level evaluation.
- Ranked first in the eligible Cognitive Track after organizer reproduction and final-layer verification; hidden-evaluation Cog score: 0.33965.

## HONORS

2026 ICPC Asia East Continent Final

Silver Medal

2025 ICPC Asia Xi'an Regional

Gold Medal

2025 CCPC Zhengzhou Site

Gold Medal

2025 ICPC Asia Shenyang Regional

Silver Medal

## TECHNICAL SKILLS

**Programming** Python, Git, testing, and reproducibility

**AI/NLP** language modeling, cognitive NLP, LLM applications, research agents, and AI for science

**Engineering** Flask, LangChain, REST APIs, Pytest, Ruff, Mypy, GitHub Actions, and LaTeX